Dana Okonkwo

Mechanical & Design Engineer

Lisbon, Portugal · dana@example.org · +351 900 000 000 · danaokonkwo.example · Portuguese citizen, no permit needed in the EU

Summary

Mechanical engineer working on single-use medical devices from first sketch to tooling release; EU patent holder. Spent three years on the clean-room side of a contract manufacturer before moving into design.

Experience

Design Engineer

Hydra Medical · Mar 2023 – Present · Lisbon

- Led the dose-counter redesign on an inhaler running 400,000 units a year, cutting the part count from nine to five and taking two weeks out of the tooling schedule.
- Built the drop and cycle test rig used for design verification, wrote the test protocols, and qualified the moulding suppliers behind them.
- Planned and ran formative usability studies (14 patients, three rounds) under IEC 62366, and fed the findings into the design history file.

Prototyping Technician

Polytechnic Prototyping Lab · Sep 2021 – Feb 2023 · Porto

- Ran a seven-machine prototyping shop: four FDM printers, a resin printer, a benchtop mill and a laser cutter.

Clean-room Technician

Medisure Contract Manufacturing · Jun 2018 – Dec 2020 · Porto

- Ran ISO 7 clean-room assembly for three catheter products, about 9,000 units a week across two shifts.
- Cut scrap on the balloon-bonding line from 6% to 2.5% by redesigning the bonding fixture.

Skills

CAD & 3D
SolidWorks (CSWP), Creo, Fusion 360, KeyShot

Fabrication
3D printing FDM/SLA, CNC milling, laser cutting, silicone tooling, injection-moulding DFM

Test & QA
ISO 13485, IEC 62366, design verification, first-article inspection, PPAP

Software
Python, MATLAB, Git, Excel

Patent

Dose-counting mechanism for a metered inhaler
CO-INVENTOR · EP 3 000 000 B1 · 2025

Education

M.Eng. Mechanical Engineering, Universidade do Porto
JUL 2021 · DISTINCTION

Languages

Portuguese: Native

English: C2

Hungarian: B1